

Melika Baghi

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Summary

Ph.D. Candidate in Industrial & Systems Engineering at Georgia Tech with an M.S. in Statistics. Work covers experiment design, ETL and feature pipelines, predictive modeling and statistical validation, through to production-oriented pipelines. Seeking data scientist and applied machine learning roles.

Technical Skills

Programming & data: Python, SQL, PySpark, R, scikit-learn, pandas, NumPy, SciPy, Airflow, MLflow

Machine learning: scikit-learn, XGBoost, PyTorch, ensemble methods, deep learning, representation and multimodal learning

Statistics: experiment design, hypothesis testing, nonparametric methods, calibration, uncertainty quantification, causal and time-series modeling

Delivery: ETL pipelines, model tracking, dashboards (Power BI), Git, Docker, Linux, GPU/HPC, cloud computing (AWS, Azure)

Projects

- **Hospital capacity and patient flow** — hybrid agent-based and discrete-event simulation of a hospital serving two patient populations. Cut the tourist queue wait **82%** (13.7 to 2.4 days) and the admission queue **43%** (25.9 to 14.8 days, 30 replications, $p < 0.001$), with a 256-run fractional factorial design over 16 operational factors. *AnyLogic, Python*
- **Predictive maintenance at scale** — end-to-end pipeline for sensor-based fault detection. ARIMA and LSTM forecasting with ensemble anomaly classifiers over multivariate telemetry gave over **15 hours** of advance warning; deployed on Spark for low-latency inference. *Python, PySpark*
- **End-to-end ML pipelines** — CRISP-DM workflows over simulation, sensor and transactional data. ETL in PySpark and SQL, model tracking automated with Airflow and MLflow, results delivered as dashboards in Python, R and Power BI.
- **Clinical risk from EHR** — adverse drug event prediction from longitudinal records, compared across regularized regression, tree ensembles and neural models, with calibration and fairness checks across demographic groups. *Python*
- **Credit risk modeling** — ensemble classifiers for consumer loan default, evaluated on ranking performance and probability calibration rather than accuracy alone. *Python*
- **Cross-national statistical inference** — nonparametric study of internet access against female employment and management representation across 92 countries; Mann–Whitney tests with Bonferroni correction and Anderson–Darling normality diagnostics. *Python*
- **Anomaly detection in sensor streams** — ARMA-based statistical monitoring with sliding-window control charts to flag faults early, trading off detection delay against false-alarm rate. *Python*
- **Breast cancer prediction** — diagnostic classification pipelines compared on held-out generalization rather than training accuracy. *Python, scikit-learn*
- **Multimodal phenotype modeling** — contrastive representation learning over single-cell transcriptomics, longitudinal EHR and imaging-derived phenotypes, with transformer language models extracting structured phenotypes from unstructured clinical notes. *Python, PyTorch*
- **Constrained gradient boosting (cXGBoost)** — enforces a manifold validity bound inside every leaf during fitting, cutting reconstruction error $5\times$ against interpolation baselines; released as an installable open-source package. [arXiv:2605.04130](#)
- **Tensor-structured surrogates** — keeps the multiway structure of simulation output instead of flattening it: $35.6\times$ – $564\times$ less interpolation work and $4\times$ – $108\times$ faster than flattened baselines across four PDE benchmarks, within 0.35 percentage points of their accuracy. Under review.
- **Prediction under missing modalities (RouteCert)** — certifies a prediction when the modalities already collected determine the decision and otherwise recommends which to acquire next, with distribution-free coverage guarantees; evaluated on ECG, speech and wearable-sensing datasets. [arXiv:2608.15520](#)

Education

Ph.D., Industrial & Systems Engineering, Georgia Institute of Technology *Aug 2023 – Dec 2027 (expected)*

M.S., Statistics, Georgia Institute of Technology *Aug 2023 – May 2026*

M.S., Industrial Engineering — Healthcare Systems, Amirkabir University of Technology; GPA 4.0/4.0 *2021 – 2023*

B.S., Industrial Engineering, Amirkabir University of Technology; GPA 3.91/4.0 *2018 – 2021*

Publications: four first-author arXiv preprints with public code ([2608.15520](#), [2609.01761](#), [2605.04130](#), [2605.19139](#)); [full list](#).